The Elements of DeFi

DeFi does not have an innovation problem. It has a vocabulary problem. Nearly every protocol ever shipped is a recombination of about fifty recurring mechanisms — and once you can name them, most of the industry stops looking novel and starts looking legible.

> Call it a periodic table if you like, but it is not one: no periodic law, no atomic number, nothing recurring down a column. Chemistry discovered its elements. These were designed, in a hurry, by people shipping to production — which is exactly why the combinations are the part worth writing down.

THEME

Dark | Light

v1.0 · reviewed 2026-08-04

48 core · 10 candidate · 10 contested · 29 laws · 19 hazards

READ A PROTOCOL

Pick one. It lights up in the table below, numbered in the order each part became necessary. Four of these are dead — and those are the instructive ones.

Aave v3
Pooled lending
10 elements

Uniswap v3
Spot exchange
4 elements

Maker / Sky
CDP stablecoin
10 elements

Liquity v1
CDP stablecoin
6 elements

GMX v2
Perpetual futures
8 elements

Lido v2
Liquid staking
5 elements

CoW Protocol
Intent exchange
4 elements

CCTP v2 Fast
Cross-domain transfer
4 elements

Centrifuge
Real-world credit
8 elements

Terra / Anchor †
Algorithmic stablecoin
6 elements

Mango Markets †
Perpetual futures
6 elements

Euler v1 †
Pooled lending
7 elements

Rows are how much must already exist

S0 needs nothing but the ledger. S4 needs other people to agree. Deeper is not more dangerous — it is more dependent, and confusing those two is how you misprice a protocol.

Blocks are families of substitutes

Everything in a block answers one question. Swap Cp for Cl and you still have a DEX. That is what a block means and the only thing it means.

The shape is the finding

Families cluster at one depth — credit is all S3, pricing all S1. And S3 is enormous: most of DeFi is machinery for owing, pricing and unwinding obligations.

Colour only ever means hazard

There is exactly one coloured thing in this interface and it is danger. Everything else earns its distinction from position, shape and lightness.

VIEW
The table | Laws | Hazards

ARRANGEMENT
By depth | By family

DIMENSION
Flat | 3D

READING ORDER
By group | By stratum | By ID

SEARCH
symbol, name or ID

Row = prerequisite depth, exactly. **Blocks** inside a row are families of substitutes. Order within a block is **not** semantic.

E001 S0 **Sh** Pro-rata share accounting ■ core	E002 S0 **Ix** Index-based accrual ■ core	E003 S0 **Rb** Rebasing accounting ■ core	CSM S1 **CSM** Constant sum degenerate limit: not an element	E004 S1 **Cp** Constant-product invariant ■ core
E005 S1 **Wg** Weighted-geometric… ■ core	E006 S1 **St** Stable-hybrid invariant ■ core	E007 S1 **Cl** Concentrated liquidity ■ core	E008 S1 **Pm** Oracle-priced inventory curve ■ core	E009 S1 **Ob** On-chain order book ■ core
E010 S1 **Rf** Request for quote ■ core	E011 S2 **Ba** Batch-auction clearing ■ core	E012 S4 **In** Intent & solver execution ■ core	E039 S1 **Ag** Aggregation & routing ■ core	E040 S1 **Fl** Atomic flash liquidity ■ core
E013 S3 **Pl** Pooled lending ■ core	E014 S3 **Im** Isolated lending market ■ core	E015 S3 **Cd** Collateralized-debt minting ■ core	E016 S3 **Uc** Undercollateralize credit ■ core	E017 S3 **Ft** Fixed-term debt ■ core
E018 S3 **Ct** Collateral-threshold test ■ core	E019 S3 **Li** Incentivized liquidation ■ core	E020 S3 **Ad** Auto-deleveraging ■ core	E021 S3 **Sl** Socialized-loss allocation ■ core	E022 S3 **Bs** Staked backstop ■ core

ID	Stratum	Symbol	Name	Status
E023	S3	**Pf**	Perpetual funding transfer	core
E024	S3	**Op**	Option payoff	core
E025	S3	**Tr**	Tranche waterfall	core
E026	S3	**Cv**	Mutual cover pool	core
E027	S3	**Py**	Principal/yield separation	core
E057	S3	**Sv**	Servicing & determination…	candidate: recurrence evidence short
E060	S3	**Dp**	Directional position & hedge…	candidate: recurrence evidence short
E028	S2	**Ex**	External data oracle	core
E029	S2	**Tp**	Time-weighted price	core
E030	S2	**Oa**	Optimistic assertion oracle	core
E031	S2	**At**	Reserve / NAV attestation	core
E032	S2	**Sr**	Streaming accrual	core
E033	S2	**Ep**	Epoch-gated transition	core
E034	S2	**Wq**	Withdrawal queue	core
E035	S2	**Em**	Protocol-funded emissions	core
E058	S2	**Fd**	Surplus & fee distribution	candidate: recurrence evidence short
E051	S3	**Gs**	Sponsored-fee liability	candidate: recurrence evidence short
E036	S4	**Tg**	Delayed-governance…	core
E037	S4	**Up**	Mutable implementation…	core
E038	S4	**Gp**	Guardian or pause	core
E050	S4	**Au**	Delegated execution scope	core
E041	S4	**Xm**	Cross-domain message…	core
E042	S4	**Xf**	Cross-domain asset transfer	core
E053	S4	**Rl**	Resource lock / reservation	candidate: recurrence evidence short
E054	S4	**Of**	Optimistic fill & reimbursement	candidate: recurrence evidence short
E043	S3	**Rd**	Direct redemption right	core
E044	S3	**Ps**	Peg-swap module	core
E045	S3	**As**	Algorithmic supply adjustment	core
E046	S2	**Aw**	Permission / identity gate	core
E047	S2	**Sb**	Shielded-balance state	core
E048	S2	**Sd**	Selective-disclosure proof	candidate: recurrence evidence short
E056	S2	**Fz**	Freeze / forced transfer	candidate: recurrence evidence short
E049	S4	**Rs**	Restaking / shared security	candidate: recurrence evidence short
E059	S3	**Vl**	Staking & validator…	candidate: recurrence evidence short

S0 Ledger-local claims 3 elements

G01 **Claims & accounting** 3

S1 Deterministic transformation 10 elements

G02 **Pool pricing** 6

G03 **Execution** 2

G04 **Liquidity catalysis** 2

S2 Measured or time-conditioned 14 elements

G03 **Execution** 1 · G08 **Truth** 4 · G09 **Time & queueing** 3 · G10 **Incentives** 2 · G14 **Access & privacy** 4

S3 Obligation & solvency 22 elements

G05 **Credit** 5 · G06 **Solvency** 5 · G07 **Risk transfer** 7 · G11 **Control & authority** 1 · G13 **Stability** 3 · G16 **Staking** 1

S4 Multi-agent & cross-domain 10 elements

G03 **Execution** 1 · G11 **Control & authority** 4 · G12 **Cross-domain** 4 · G15 **Security reuse** 1

Contested register — 10 entries

Held outside the table because a necessary criterion is genuinely disputed. Not usable in a formula without a note. The separation means “argued about”, not “filed away”.

P001
Bc
Bonding-curve issuance
n contested: note required

P002
Tw
Time-weighted AMM execution
n contested: note required

P003
Da
Dutch-auction descent
n contested: note required

P004
Cg
Credit delegation
n contested: note required

P005
Ir
Insurance reserve fund
n contested: note required

P006
Zk
Verifiable state proof
n contested: note required

P007
Ve
Vote-escrow allocation
n contested: note required

P008
Kg
Credential-gated transfer
n contested: note required

P009
Ua
Unified-balance ledger
n contested: note required

P010

Sq

Shared ordering commitment

contested: note required

HOW TO THINK WITH IT

Six questions, in this order. Run them against anything — a protocol you are auditing, a pitch deck, your own design — and the table stops being a diagram and becomes a method. Question three pays for the other five.

1

What are its parts?

Name every recurring financial state transition and refuse everything else. ERC-4626 is a socket, not a mechanism. A hook is somewhere to put an element, not an element. “Points” is marketing. Most whitepapers describe three real mechanisms and forty pages of scenery.

2

What did each part force?

Parts are not chosen freely — each one drags in the next. Leverage forces a truth source, a solvency test and a way for the position to end. A protocol with leverage and only two of those three is not innovating. It is missing a part, and you have just found it.

3

Where does it read the outside world?

This is the question that pays. Every catastrophic loss this table explains runs through the price, not the code. Ask what it would cost to move that input — measured against pool depth, never against the attacker’s balance — and if the answer is less than what can be borrowed against it, the protocol is already lost.

4

Does anything feed itself?

Reflexive collateral is the most reliable way to lose everything ever deployed on-chain. A token that is simultaneously the collateral, the market that prices it, and the capital backing it has three defences that all fail on the same news. This is the one shape that goes to zero rather than merely down.

5

Who eats the loss?

For every fee, delay, slippage and advance, name the claim class or the underwriter. An unassigned loss has not vanished — somebody is carrying it without being paid to, and usually it is the depositor who never read this far. If the documentation cannot name them, that is the answer.

6

What did the formula fail to say?

The residue, and it is the most useful line on the page. For a lending market it is small. For anything touching real-world assets it is larger than the formula and contains the actual risk. A decomposition that reports no residue was not finished.

LEGEND

Stratum S0 → S4

Prerequisite depth, not risk. Carried by the band, the S-token and lightness — never hue.

- core
- candidate: recurrence evidence short
- contested: note required

- hazard: forbidden
- hazard: elevated
- hazard: unverifiable

degenerate limit: not an element

no denominator — failures counted, survivors never were

Hazard membership is categorical. It is never a probability, because no exposure or survivor counts exist.

—i→ Interface — Can these exchange the required calls, tokens, messages, proofs?

—e→ Economic — Do combined incentives, liquidity and loss allocation stay solvent?

—t→ Trust — Who can attest, censor, upgrade, pause, mint, freeze?

—n→ Informational — Who observes the order before it settles?

This table explains roughly two thirds of DeFi incidents by count and a minority of them by dollars. That is not a defect in the table — it is the most useful finding on this page. Your largest risk is not your financial design. It is your keys, your signing interface and your build pipeline. Get the mechanism right and you have eliminated the failures people write papers about, not the ones that actually took the money.